
Table of Contents

1. Game Overview	3
2. Core Game Loop	3
3. Gacha Pull System	4
3.1 Banner Types	5
3.2 Pull Rates	5
3.3 Pull Costs	5
3.4 Pity System	6
3.5 Duplicate Handling	6
4. Studio Room Management	6
4.1 Station Types	7
4.2 Station Upgrades	7
4.3 Resource Output Calculation	8
5. Currency System	8
6. Character Progression	9
6.1 Character Levelling	9
6.2 Character Ascension	10
7. Studio Level and Feature Unlocks	10
7.1 Odekake (Going Out) - Planned Feature	11
8. Collection Gallery	11
9. UI/UX Design	12
9.1 Navigation	12
9.2 Card Design	13
9.3 Pull Animation	13
9.4 Responsive Design	13
10. Super Chat Toss Minigame	14

11. Stamina System	14
12. Technical Architecture	15
12.1 Technology Stack	15
12.2 File Structure	15
12.3 Save/Load System	16
12.4 Game Tick System	16
13. Development Status and Sprint Plan	17
14. Character Data	18
14.1 Agency Distribution	18
15. Appendix: Key Data Structures	19

1. Game Overview

MyVT Gacha Collection is a browser-based idle gacha collection game themed around the Malaysian VTuber (MyVT) community. Players collect virtual cards featuring real Malaysian VTubers by pulling from gacha banners, then manage a virtual studio where assigned VTubers generate resources passively. Built entirely with vanilla HTML, CSS, and JavaScript, the game is hosted on GitHub Pages and requires no installation, no backend server, and no database. All game state is persisted using the browser's localStorage, making it a truly zero-friction experience that players can enjoy on any device with a modern browser.

The roster features 319 Malaysian VTubers sourced from hololist.net, spanning independent creators and members of various agencies including VGakuenLive, HoloDream, MyHolo TV (VILIT), Projek Hikayat, and Phase Connect. Each VTuber is represented by a character card with portrait art from their hololist entry, and cards can be pulled in three variant tiers: Normal, SR (Super Rare), and SSR (Super Super Rare). The variant tier determines the card's visual frame, resource output multiplier, and level cap, creating a compelling collection and progression system built around the diverse Malaysian VTuber community.

The game design is guided by five core pillars: Accessibility, meaning zero install and any-browser compatibility; Community Celebration, showcasing the vibrant MyVT scene to a wider audience; Idle-Friendliness, where progress continues even when the player is offline; Collection Satisfaction, with 319 characters and variant upgrades providing long-term goals; and Skill Development, as the project serves as a practical learning exercise for web game development using vanilla technologies.

The primary target audience consists of members of the MyVT community, including VTubers themselves, their fans, and supporters of the Malaysian VTuber scene. Secondary audience members include general gacha game enthusiasts and idle game players who may discover the game through community sharing. The game is designed to be approachable for newcomers while offering enough depth to retain experienced gacha players. All text in the game is presented in English, reflecting the multilingual nature of the Malaysian VTuber community.

2. Core Game Loop

The core gameplay loop of MyVT Gacha Collection revolves around a seven-step cycle that intertwines gacha pulling, idle resource generation, and character progression. Players begin by spending Stars on gacha pulls to acquire VTuber character cards. These cards are then assigned to Studio Room stations, where

they passively generate resources even when the player is offline. Accumulated resources are reinvested to level up characters, ascend them to higher variant tiers, upgrade studio stations, and ultimately unlock new features and stations. The loop is designed to be simple enough to understand in seconds but deep enough to sustain engagement over weeks and months of regular play, with each step feeding naturally into the next.

New players start with 1,000 Stars, enough for approximately 10 single pulls on the Standard Banner. The first few pulls will typically yield Normal variant characters, which the player can immediately assign to the Stream Room station to begin generating Stars passively. This establishes the resource generation foundation that fuels the rest of the gameplay loop. As players progress, they unlock additional station types, each producing different currencies required for character progression, creating a web of interdependencies that keeps all game systems relevant throughout the play experience.

Step	Action	Description
1	Pull	Spend Stars on gacha banners to acquire VTuber character cards.
2	Assign	Place collected characters into Studio Room stations to generate resources.
3	Earn	Resources accumulate passively (idle), even while the player is offline.
4	Level Up	Spend Star Dust and Stars to increase character levels, boosting station output.
5	Ascend	Spend Star Fragments to upgrade Normal to SR and SR to SSR variants.
6	Expand	Upgrade Studio Level to unlock new stations, slots, and features.
7	Repeat	Continue pulling to fill collection gaps; aim for 100% SSR completion.

Table 1: The Seven-Step Core Game Loop

3. Gacha Pull System

The gacha pull system is the primary mechanism through which players acquire new character cards. It features two banner types, three variant tiers with distinct drop rates, a pity system to prevent excessive bad luck, and a duplicate handling system that ensures every pull produces meaningful value. The system is designed to create excitement and anticipation with each pull while maintaining fairness and long-term progression for dedicated players.

3.1 Banner Types

The gacha system features two types of banners. The Standard Banner contains all 319 Malaysian VTubers from the hololist roster and is permanently available, serving as the primary source of new characters. The Featured Banner is a rotating limited-time banner that highlights 3-4 specific VTubers with increased drop rates on a weekly rotation. Both banners draw from the full 319-character pool for character selection, meaning that any character can be pulled from either banner. The Featured Banner's rate-up characters simply have a higher probability of appearing, making it the preferred choice for players seeking specific VTubers for their collection.

Banner Type	Pool Size	Featured Characters	Rotation
Standard	319	None (equal rates)	Permanent
Featured	319	3-4 rate-up VTubers	Weekly

Table 2: Banner Types Overview

3.2 Pull Rates

Each pull yields one character card with a randomly determined variant tier. The variant system does not affect which character is obtained; rather, it determines the rarity and visual quality of that character's card. Normal cards appear at a base rate of 82%, SR cards at 15%, and SSR cards at 3%. This means that on average, a player will see an SSR card roughly once every 33 pulls, though the pity system ensures that no player goes more than 50 pulls without an SSR. The 10x multi-pull additionally guarantees at least one SR or above card, providing a reliable floor for multi-pull investments.

Variant	Drop Rate	Card Frame	Pull Guarantee
Normal	82%	Grey border, standard art	Always obtainable
SR	15%	Silver border with glow	1 per 10x pull
SSR	3%	Gold border with shine	Pity at 50 pulls

Table 3: Pull Rates by Variant Tier

3.3 Pull Costs

Pulling from either banner costs Stars, the game's primary currency. A single pull costs 100 Stars, while a ten-pull costs 1,000 Stars (equivalent to 10 single pulls with no discount). The pricing is intentionally simple and consistent across both

banner types, ensuring that players can easily calculate how many pulls they can afford at any given time. The starting bonus of 1,000 Stars provides enough for either 10 single pulls or one multi-pull, giving new players an immediate and meaningful introduction to the gacha mechanic.

Pull Type	Cost (Stars)	Guarantee
Single Pull (1x)	100	None
Multi Pull (10x)	1,000	At least 1 SR or above

Table 4: Pull Costs

3.4 Pity System

A pity system ensures that players are guaranteed an SSR variant after 50 consecutive pulls without obtaining one. The pity counter tracks pulls across all banners and resets every time an SSR variant is pulled, regardless of which banner it came from. This cross-banner tracking means that players cannot accidentally lose their pity progress by switching banners, and it prevents excessively unlucky streaks from undermining the player experience. Additionally, the 10x multi-pull guarantees at least one SR or above card, providing a reliable floor for multi-pull investments and ensuring that even unlucky multi-pulls yield at least one notable result.

3.5 Duplicate Handling

When a player pulls a character they already own at the same variant tier, the duplicate is converted into Shards. Shards serve as a universal resource that can be exchanged for Star Dust or Star Fragments at a conversion rate displayed in the in-game interface. If a player already owns a character at SSR tier and pulls that same character at any tier again, the duplicate is automatically converted into bonus Stars instead of Shards, providing a meaningful endgame resource sink. This duplicate handling system ensures that every pull produces value, even when the character pool has been largely exhausted, which is especially important given the relatively small roster of 319 characters.

4. Studio Room Management

The Studio Room is the management hub of MyVT Gacha Collection, providing the idle resource generation layer that transforms the game from a simple pull simulator into an engaging management experience. Players assign their collected VTuber characters to various Stations, each of which produces a

different type of resource. The strategic decision of which characters to assign to which stations, combined with station upgrades and character progression, creates a deep and satisfying management loop that rewards thoughtful planning and long-term engagement.

4.1 Station Types

Each station represents a different activity that VTubers might engage in, and each produces a unique resource. Players start with the Stream Room station unlocked at Studio Level 1 and gradually unlock additional stations by increasing their Studio Level. Every station can hold exactly one assigned VTuber at a time, and the player may reassign characters freely with no cooldown or penalty. The output of each station scales with the assigned character's variant tier and level, as well as the station's own upgrade level, creating multiple axes of optimization for players to explore.

Station	Resource	Primary Use	Unlock
Stream Room	Stars	Gacha pulls, station upgrades	Studio Lv 1
Creative Corner	Star Dust	Character levelling	Studio Lv 2
Practice Hall	Star Fragments	Ascension material	Studio Lv 4
Lounge	Bond Points	Studio Level EXP	Studio Lv 6

Table 5: Station Types and Their Resources

4.2 Station Upgrades

Each station can be upgraded from Level 1 through Level 5. Upgrading increases the station's output multiplier, meaning that the same assigned character will produce more resources per minute at higher station levels. The output multiplier progresses as follows: Level 1 provides a x1 multiplier, Level 2 provides x1.5, Level 3 provides x2, Level 4 provides x3, and Level 5 provides x5. This exponential scaling means that maxing out a station is a significant long-term investment that dramatically increases resource generation, and the scaling costs ensure that players must carefully prioritise which stations to upgrade first.

Level	Multiplier	Cost (Stars)	Cost (Fragments)
1	x1	(Base)	(Base)
2	x1.5	500	50

3	x2	2,000	200
4	x3	8,000	800
5	x5	30,000	3,000

Table 6: Station Upgrade Costs and Multipliers

4.3 Resource Output Calculation

The resource output per minute for any station is determined by a multiplicative formula: $\text{Output} = \text{Base Rate} \times \text{Character Variant Multiplier} \times \text{Character Level Bonus} \times \text{Station Level Multiplier}$. The base rate differs per station type. Character variant multipliers are x1 for Normal, x2 for SR, and x5 for SSR. The character level bonus is calculated as $1 + (\text{Level} \times 0.02)$, meaning a Level 50 SSR character at a Level 5 station produces $(\text{Base Rate} \times 5 \times 2 \times 5) = 50x$ the base rate. This formula ensures that every aspect of character progression meaningfully contributes to resource generation, making both pulling higher-tier cards and investing in levelling feel rewarding.

5. Currency System

MyVT Gacha Collection features five distinct currencies, each serving a unique purpose within the game economy. This multi-currency design ensures that players must engage with all aspects of the game to progress, preventing the common gacha game problem of a single currency becoming irrelevant after the early game. Four of the five currencies are produced by studio stations, while the fifth, Gems, is a premium currency earned primarily through the Super Chat Toss minigame and collection milestones. Each currency feeds into different progression systems, creating interdependencies that make the gameplay loop feel cohesive and rewarding throughout the entire play experience.

Currency	Sy mb ol	Source	Primary Use
Stars	*	Stream Room, Daily Login, Duplicate SSR	Gacha pulls, Station upgrades
Star Dust	+	Creative Corner, Shard conversion	Character levelling
Star Fragments	#	Practice Hall, Shard conversion	Ascension crafting

Bond Points	~	Lounge station, Odekake outings	Studio Level EXP
Gems	\$	Minigame, Milestones	Premium rewards, exchange

Table 7: Currency Types, Sources, and Uses

New players receive a starting package of 1,000 Stars to immediately begin pulling on banners. No other currencies are provided at the start; players must progress through the early game to unlock stations and begin generating secondary currencies. The Daily Login Bonus provides 100 Stars as a base amount, increasing by 50 Stars per consecutive day of login, with a daily cap of 500 Stars after a 9-day streak. Missing a day resets the streak to day 1, creating a gentle but meaningful incentive for regular engagement. Gems, the premium currency, are earned by playing the Super Chat Toss minigame and by reaching collection milestones, providing an additional progression path beyond passive resource generation.

6. Character Progression

Characters in MyVT Gacha Collection progress through two interconnected systems: levelling and ascension. Levelling strengthens a character within its current variant tier, increasing its resource output when assigned to a station. Ascension upgrades a character to a higher variant tier (Normal to SR, or SR to SSR), unlocking a new card frame, higher level caps, and significantly increased output multipliers. Both systems require different currencies, ensuring that players must engage with multiple station types to fully develop their roster. The progression system is designed to provide constant short-term goals while maintaining compelling long-term targets for dedicated players.

6.1 Character Levelling

Each character can be levelled from Level 1 up to a maximum level determined by their current variant tier. Normal variant characters have a level cap of 20, SR characters can reach Level 35, and SSR characters can reach Level 50. Levelling requires spending Star Dust and Stars, with costs scaling based on the target level. Each level increases the character's resource output by approximately 2% (the level bonus formula is $1 + \text{Level} \times 0.02$), making levelling a consistent and predictable investment that pays dividends across all station types where the character might be assigned.

Variant	Level Cap	Cost (Star Dust)	Cost (Stars)
---------	-----------	------------------	--------------

Normal	20	$10 + (\text{Lvl} \times 5)$	$20 + (\text{Lvl} \times 10)$
SR	35	$30 + (\text{Lvl} \times 10)$	$50 + (\text{Lvl} \times 20)$
SSR	50	$80 + (\text{Lvl} \times 20)$	$100 + (\text{Lvl} \times 40)$

Table 8: Levelling Costs by Variant Tier

6.2 Character Ascension

Ascension is the process of upgrading a character to a higher variant tier. To ascend a Normal character to SR, the character must be at maximum level (20) and the player must spend 100 Star Fragments. To ascend an SR character to SSR, the character must be at its maximum SR level (35) and the player must spend 500 Star Fragments. Ascension changes the character's card frame, increases the variant output multiplier (from x1 to x2 for Normal-to-SR, and from x2 to x5 for SR-to-SSR), raises the level cap, and increases Studio EXP contribution. This dual-gate system (max level + fragment cost) ensures that ascension is a significant milestone that requires sustained investment in each individual character.

Ascension	Required Level	Cost (Fragments)	New Level Cap
Normal to SR	Lv 20	100	35
SR to SSR	Lv 35	500	50

Table 9: Ascension Requirements

7. Studio Level and Feature Unlocks

The Studio Level serves as the overarching progression gate for the game. Unlike character levels or station upgrades, the Studio Level does not directly improve resource generation rates. Instead, it unlocks new features, stations, and station slots, providing a structured sense of progression that guides players through the game's systems at a measured pace. Studio EXP is earned passively from assigned characters in stations, with SSR characters contributing more EXP than SR, and SR contributing more than Normal. The Studio Level ensures that new players are not overwhelmed by too many systems at once while providing experienced players with clear goals to work toward over the course of their playthrough.

Studio Lv	Unlocks	EXP Required
-----------	---------	--------------

1	Stream Room + 2 station slots	0
2	Creative Corner station	200
3	3rd station slot	600
4	Practice Hall station	1,500
5	4th station slot	3,500
6	Lounge station	7,000
7	Character detail stats	12,000
8	5th station slot	20,000
9	Agency filter	35,000
10	Odekake (Going Out)	60,000

Table 10: Studio Level Progression and Unlocks

7.1 Odekake (Going Out) - Planned Feature

At Studio Level 10, players unlock the Odekake feature, inspired by the training camp system in Uma Musume. Odekake allows players to send a character on a timed "outing" lasting between 1 and 4 hours. During this time, the character cannot be assigned to any station. When the outing concludes, the character returns with Bond Points, Studio EXP, and occasionally rare crafting materials. Repeated outings with the same character build a Bond Level, which unlocks character-specific bonuses such as increased output multipliers or reduced ascension costs. This feature adds an emotional attachment layer to the game, encouraging players to invest in their favourite characters beyond mere numerical optimization. The Bond Level system provides a long-term progression goal that extends beyond the collection and variant upgrade mechanics, giving players a reason to continue engaging with the game even after achieving a high collection completion percentage.

8. Collection Gallery

The Collection Gallery is the player's visual archive of all 319 Malaysian VTubers in the game. It displays every character as a card in a responsive grid layout, with distinct visual indicators showing which characters the player owns, which variant tiers they have obtained, and which characters remain undiscovered. The gallery serves as both a progress tracker and a showcase of the player's collection achievements, providing a clear visual representation of how far the player has come and what remains to be collected.

The gallery provides multiple filter options to help players navigate the large roster. Ownership filters allow viewing All, Owned, or Unowned characters. Variant filters narrow results by highest owned variant (Normal, SR, or SSR). An Agency filter, unlocked at Studio Level 9, enables filtering by agency affiliation such as Independent, VGakuenLive, VILIT, and others. A Station filter shows characters by assignment status (Any, Currently Assigned, or Unassigned). Players can also search by VTuber name and sort by Name or Variant tier. The gallery displays 40 characters per page with pagination controls.

A collection progress bar at the top of the gallery shows overall completion percentage alongside per-variant completion stats (for example, 45/319 SSR, 120/319 SR, 280/319 Normal). Collection milestones at 10, 25, 50, 100, 150, 200, 250, 300, and 319 owned characters reward players with bonus resources and Gems, providing structured goals and celebration moments throughout the collection journey. Clicking any character card opens a detail popup showing the character's pull history, current stats, all owned variants, and variant-specific information.

9. UI/UX Design

The game's user interface is built around a dark theme that follows the visual language common in anime-style gacha games, creating an immersive and familiar aesthetic for the target audience. The primary background is a deep dark grey (#0e0e0c), with card backgrounds in slightly lighter tones to create visual depth. Purple accents (#896fd9) are used for interactive elements and headers, teal/green accents (#5dd499) highlight positive feedback such as resource gains, and gold tones are reserved for SSR-tier elements to convey rarity and premium quality.

9.1 Navigation

The user interface employs a tab-based navigation system with five primary sections: Home, Pull, Collection, Studio, and Minigame. A persistent top navigation bar displays these tabs along with the player's current resource counts (Stars, Star Dust, Star Fragments, Bond Points) and a stamina bar. A settings overlay accessible from the navigation bar provides save, export, import, device transfer, and reset functionality. The navigation is designed to be thumb-friendly on mobile devices, with tabs positioned at the bottom on narrow viewports and at the top on wider screens.

Tab	Content	Key Interactions
-----	---------	------------------

Home	Stats, daily login, offline earnings, studio overview	Claim offline earnings, view progress
Pull	Banner selection, pull animation, results	Single/multi pull, banner switching
Collection	Character gallery grid with filters	Filter, sort, view character details
Studio	Stations, assignment, upgrades	Assign characters, upgrade stations
Minigame	Super Chat Toss game screen	Play rounds, view high scores

Table 11: Navigation Tabs and Their Content

9.2 Card Design

Character cards are the central visual element of the game. Each card displays the VTuber's portrait image from hololist.net, their name, agency, and variant tier badge. Normal cards have a simple grey border, SR cards feature a silver border with a subtle glow effect, and SSR cards display a gold border with an animated shine effect. Cards are sized at approximately 120x160 pixels in the gallery grid and scale up to 200x280 pixels in single-pull reveal and 120x168 pixels in 10-pull reveal. The pull animation features a sequential slide-in with a 3D flip reveal, where cards appear one at a time from below the screen and flip to reveal the character. Single pulls take approximately 1 second for a dramatic reveal, while 10-pulls complete in about 8 seconds total with an option to tap and skip.

9.3 Pull Animation

The pull animation system employs a sequential slide-in combined with a 3D card flip. Cards appear one at a time from below the viewport, slide up to the centre, and flip to reveal the character. For single pulls, this takes approximately 1 second with a dramatic pause before the flip. For 10-pulls, the total animation runs for about 8 seconds with cards appearing in rapid succession. Players can tap to skip the animation at any point. SR pulls trigger a silver screen flash effect, and SSR pulls trigger a golden screen flash accompanied by an animated golden glow ring around the card. These visual effects create moments of excitement and celebration that reinforce the emotional reward of pulling higher-tier cards.

9.4 Responsive Design

The game is designed to be fully responsive, adapting to screen sizes from 320px mobile screens to 1920px desktop displays. On mobile, the gallery grid displays 3 cards per row with a bottom navigation bar. On tablet, the grid expands to 4-5 cards per row. On desktop, up to 8 cards per row are shown with a horizontal top

navigation bar. The Studio Room layout adapts similarly, showing stations in a vertical stack on mobile and a grid on desktop. All interactive elements have touch targets of at least 44x44 pixels for comfortable mobile interaction.

10. Super Chat Toss Minigame

The Super Chat Toss is a tap-based 30-second minigame accessible from the Minigame navigation tab. It provides an active gameplay alternative to the idle resource generation of the studio system, offering players a skill-based way to earn Stars and Gems. The minigame features five bubble types that float across the screen: Coin bubbles grant +10 Stars, Premium bubbles grant +25 Stars, Gem bubbles grant +1 Gem, Landmine bubbles penalise the player with -15% of their current score, and Gift bubbles add +3 seconds to the remaining time, extending the play session.

A boost meter fills as the player taps bubbles. When the meter reaches 100%, it triggers Super Chat Mode for 5 seconds, during which all Coin and Premium bubbles are worth 5 times their normal value. This creates exciting moments of high-scoring potential that reward sustained accuracy and quick reflexes. Before each round, the player selects an owned VTuber as their "streamer lead," whose variant tier determines a score multiplier: Normal characters provide x1, SR characters provide x1.5, and SSR characters provide x2. This character lead system adds a strategic layer, encouraging players to invest in and use their highest-tier characters during minigame rounds.

Each round of Super Chat Toss costs 15 stamina points. The results screen at the end of each round displays the Stars earned, Gems earned, and Bond Points gained (+10 per round). High scores are tracked locally, giving players a personal best to beat. The minigame serves multiple design purposes: it provides active engagement for players who want more than idle gameplay, it creates a meaningful use for stamina, it offers a path to earn the premium Gem currency, and it gives owned characters additional utility beyond studio station assignment.

11. Stamina System

The stamina system governs access to the Super Chat Toss minigame, adding a resource management layer to the active gameplay experience. Players have a maximum of 200 stamina points, which regenerate at a rate of 1 point every 4 minutes (approximately 360 points per day at full regeneration). Each round of Super Chat Toss costs 15 stamina, meaning a fully charged stamina bar allows for

approximately 13 consecutive rounds of play before the player must wait for regeneration.

The stamina bar is prominently displayed in both the navigation bar and the minigame start screen, alongside a timer showing the time until the next point of recovery. When the player loads the game after being offline, the system calculates offline recovery based on the elapsed time since the last save, ensuring that players do not lose any regeneration progress during breaks. This offline recovery calculation is capped to prevent excessive accumulation during very long absences. The stamina system creates a natural cadence for minigame engagement, preventing the minigame from becoming the sole focus of gameplay while still providing regular opportunities for active play sessions.

12. Technical Architecture

The game is built entirely with vanilla web technologies: HTML5 for structure, CSS3 for styling and animations, and JavaScript ES6+ for game logic. No frameworks, libraries, or build tools are required, making the project accessible to developers of all skill levels and easy to deploy on any static hosting platform. The only external dependency is the character portrait images hosted on hololist.net, which are loaded dynamically via URLs stored in the game's character data file. This zero-dependency approach was chosen deliberately to keep the project simple, educational, and easy to maintain.

12.1 Technology Stack

Technology	Purpose	Rationale
HTML5	Page structure	Universal browser support
CSS3	Styling, animations, responsive layout	Card flip, dark theme, media queries
JavaScript ES6+	Game logic, state, DOM	No framework overhead
localStorage	Client-side persistence	No backend, instant save/load
JSON	Character roster data	Human-readable, easy updates

Table 12: Technology Stack

12.2 File Structure

The project follows a clean, modular file structure that separates concerns for maintainability and readability. Each JavaScript file handles a distinct area of game functionality, and the character data is stored in a separate JSON file that can be updated independently. Note that the original GDD listed `js/studio.js` and `js/firebase.js`, which do not exist in the current codebase. Studio logic is handled within `game.js`, and Firebase cloud save has not been implemented beyond placeholder UI buttons in the settings modal.

File	Purpose
<code>index.html</code>	Main page, loads all scripts
<code>css/style.css</code>	All styles: dark theme, animations, responsive
<code>js/data.js</code>	Character data loader (fetches <code>characters.json</code>)
<code>js/game.js</code>	Core state, save/load, tick, currencies, stamina, studio, milestones
<code>js/gacha.js</code>	Pull logic, rates, pity, featured banner, Fisher-Yates shuffle
<code>js/characters.js</code>	Levelling, ascension, shards, variant management
<code>js/minigame.js</code>	Super Chat Toss engine (bubbles, boost, scoring)
<code>js/ui.js</code>	UI rendering, navigation, pull animation, gallery, studio UI, toasts
<code>data/characters.json</code>	319 VTuber entries (name, slug, image, agency, url)

Table 13: Project File Structure

12.3 Save/Load System

All game state is persisted to the browser's `localStorage` under a single JSON key. The game auto-saves every 30 seconds and on every significant player action, ensuring minimal data loss. Players can also manually save via a button in the settings overlay. Export and import functionality allows players to transfer save data between devices using Base64-encoded save codes. A migration system handles save version updates, automatically converting old save data to the latest format when the game loads. Error resilience is built into the initialisation process: a try-catch block on startup catches any corruption or compatibility issues and displays an error message directing the player to check the browser console, rather than showing an infinite "Loading..." screen.

12.4 Game Tick System

A 1-second interval game tick runs whenever the game is open, updating resource generation, stamina recovery, and the user interface. When the player closes the

game and returns later, offline earnings are calculated based on the elapsed time since the last save, with a cap of 12 hours of offline resource accumulation. This ensures that extended offline periods do not result in excessively large resource windfalls that could unbalance the game economy. The tick system also handles cooldown timers, UI refreshes, and any time-dependent game events.

13. Development Status and Sprint Plan

The original 9-sprint development plan has been partially completed with significant scope changes. The first four sprints (Core UI Shell, Gacha Pull System, Collection Gallery, and Studio Room) have been fully completed, establishing the core gameplay loop. Sprint 5 was reimplemented as Profile and History (originally planned as Currency and Idle System, but most of that functionality was integrated into earlier sprints). Sprint 8 (Super Chat Toss) was pulled forward and completed, displacing the originally planned Polish and Save/Load sprint. The remaining sprints show varying levels of completion.

Sprint	Focus	Status
1	Core UI Shell	Completed
2	Gacha Pull System	Completed
3	Collection Gallery	Completed
4	Studio Room	Completed
5	Profile and History	Completed / Partial
6	Character Progression	Partial
7	Studio Progression	Partial
8	Super Chat Toss	Completed
9	Cloud Save Sync	Not Started

Table 14: Sprint Status Overview

Several code review items have been deferred to future sprints. High-priority deferred items include XSS vulnerabilities in user-facing text (HP-03) and deprecated API usage (HP-05). Medium-priority items include an auto-backup system (V4-08). Low-priority polish items (LP-01 through LP-05) include various UI refinements and edge-case handling improvements. These items are

documented in the code review tracker and will be addressed in a dedicated code quality sprint.

Proposed future sprints include Sprint 10 (Code Quality and Security), Sprint 11 (Odekake Going Out feature requiring Studio Level 10), Sprint 12 (Social Features including leaderboards and friend lists), and Sprint 13 (Content Updates with new VTuber additions, seasonal events, and limited-time banners). These sprints represent the long-term development roadmap beyond the initial release, extending the game's content and feature set based on player feedback and community engagement.

14. Character Data

The character roster is sourced from hololist.net, which maintains a comprehensive database of Malaysian VTubers. Each character entry in the data file (`characters.json`) contains five fields: name, slug, image URL, agency, and profile URL. The slug serves as a unique identifier used for internal game logic and save data references, while the image URL points to the character's portrait on hololist.net. The agency field is used for filtering in the Collection Gallery (unlocked at Studio Level 9) and for display on character cards.

14.1 Agency Distribution

The 319-character roster is heavily weighted toward independent creators, who make up approximately 79.6% of the total roster. This distribution reflects the current state of the Malaysian VTuber scene, where independent creators form the overwhelming majority. The largest agency represented is VGakuenLive with 7 members (2.2%), followed by HoloDream with 6 members (1.9%), Slice of Six Crew with 5 members (1.6%), and AmaiUsagi with 4 members (1.3%). The remaining 43 characters (13.5%) belong to various smaller agencies and groups.

Agency	Count	Percentage
Independent	254	79.6%
VGakuenLive	7	2.2%
HoloDream	6	1.9%
Slice of Six Crew	5	1.6%
AmaiUsagi	4	1.3%
Others (combined)	43	13.5%

Total	319	100%
--------------	------------	-------------

Table 15: Agency Distribution

15. Appendix: Key Data Structures

The game state is stored as a single JSON object in localStorage. Below is the current state structure showing all top-level fields and their types. This reference is intended for developers working on the codebase who need to understand the save data format. The structure includes fields for player resources, character collection, studio configuration, gacha state, minigame statistics, stamina, milestone tracking, and pull history.

The top-level game state object contains the following fields: **stars** (number), **starDust** (number), **starFragments** (number), **bondPoints** (number), **gems** (number), **characters** (object keyed by slug, each with level, variant, shards, and ascension status), **studio** (object with level, exp, station assignments and upgrade levels), **gacha** (object with pityCounter and lastPullTimestamp), **stamina** (object with current value and lastRecoveryTimestamp), **minigame** (object with highScore, totalRoundsPlayed, and totalStarsEarned), **milestones** (array of claimed milestone IDs), **pullHistory** (array of pull records with timestamp, banner, and results), **lastSave** (ISO timestamp), **dailyLogin** (object with current streak and lastClaimDate), and **version** (string indicating save data schema version).

The migration system uses the version field to detect when save data needs to be upgraded. When a player loads the game with an older save format, the migration code transforms the data to the latest schema before initialising the game state. This approach ensures backward compatibility and allows the save format to evolve across releases without requiring players to manually reset their progress.